

5. изг. Gamma распр. параметров на $[0, 2\theta]$

$$Z \sim R(0, 2\theta)$$

$$p(x, \theta) = \frac{1}{\theta} \mathbb{I}_{(0, 2\theta)}(x)$$

$$\alpha_k = \int_{-\infty}^{+\infty} x^k df(x, \theta) = \alpha_k(\theta)$$

$$M\{Z\} = \alpha_1 = \int_0^{2\theta} x \cdot \frac{1}{\theta} dx = \frac{x^2}{2} \cdot \frac{1}{\theta} \Big|_0^{2\theta} = \frac{4\theta^2}{2} \cdot \frac{1}{\theta} - \frac{0^2}{2} \cdot \frac{1}{\theta} = \frac{3}{2}\theta$$

$$M\{Z^2\} = \alpha_2 = \int_0^{2\theta} \frac{x^2}{\theta} dx = \frac{7\theta^2}{3}$$

$$D\{Z\} = \mu_2 = M\{Z^2\} - M^2\{Z\} = \frac{7\theta^2}{3} - \frac{3}{4}\theta^2 = \frac{\theta^2}{12}$$

ОММ: $\alpha_1(\theta) = \tilde{\alpha}_1$

$$\frac{3}{2}\theta = \bar{x} \Rightarrow \tilde{\theta}_1 = \frac{2}{3}\bar{x}$$

• не смещенность: $M\tilde{\theta}_1 = \theta$: $M\tilde{\theta}_1 = M(\frac{2}{3}\bar{x}) = \frac{2}{3}M\{Z\} = \theta \Rightarrow \tilde{\theta}_1$ - не смещ.

• состоятельность: $D\tilde{\theta}_1 = \frac{4}{9n} D\{Z\} = \frac{4\theta^2}{108n}$
 $= \frac{\theta^2}{27n} \xrightarrow{n \rightarrow \infty} 0 \xRightarrow{\text{по лем. Вар.}} \tilde{\theta}_1$ - конс.

$$\text{OMN: } L(\theta) = \prod_{i=1}^n f(x_i, \theta) =$$

$$\Rightarrow L(\theta) = \frac{1}{\theta^n} \{ (\theta < x_i < 2\theta) \}_{i=1}^n,$$

$$x_{\max} < 2\theta$$

$$x_{\min} > \theta$$

$$\Rightarrow \hat{\theta}_2 = \frac{x_{\max}}{2}$$

$$\Rightarrow \theta \in \left(\frac{x_{\min}}{2}; x_{\min} \right)$$

$$\bullet \text{ неслучайность: } M\hat{\theta}_2 = \theta$$

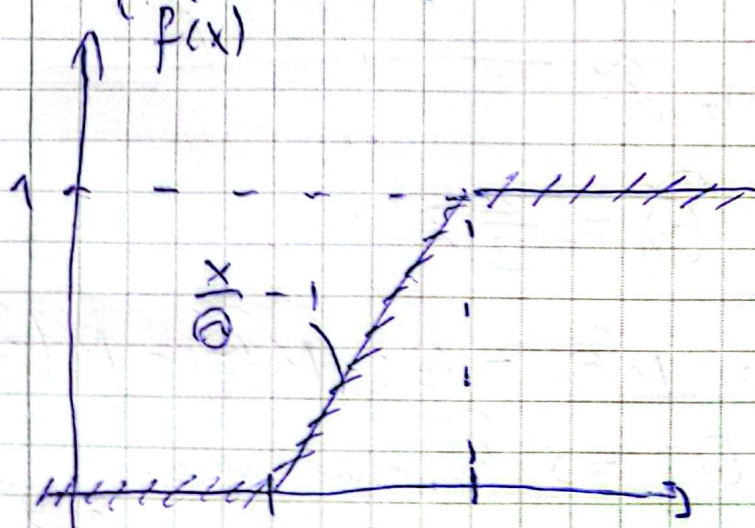
$$M\hat{\theta}_2 = M \frac{x_{\max}}{2}$$

$$x_{\max} \sim \psi(x) = (f(x))^{\frac{1}{n}}$$

$$\psi(x) = \psi'(x) =$$

$$= n \cdot \left(\frac{x}{\theta} - 1 \right)^{n-1} \cdot \frac{1}{\theta}$$

$$\cdot \{ (\theta, 2\theta) \}$$

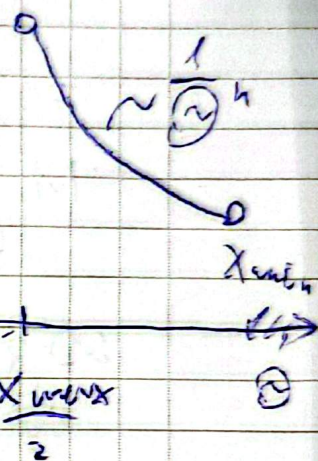


$$M \frac{x_{\max}}{2} = \frac{1}{2} \int_{\theta}^{2\theta} x \left(\frac{x}{\theta} - 1 \right)^{n-1} \frac{1}{\theta} dx =$$

$$= \left\{ t = \frac{x}{\theta} - 1 \right\} =$$

$$= \frac{n\theta}{2} \int_0^1 (1+t) t^{n-1} dt = \frac{n\theta}{2} \left(\frac{1}{n} + \frac{1}{n+1} \right)$$

$$= \frac{n\theta}{2} \cdot \frac{(2n+1)}{n(n+1)} = \frac{\theta}{2} \cdot \frac{(2n+1)}{(n+1)}$$



$\Rightarrow \tilde{\Theta}_2$ - анрег.

$$\tilde{\Theta}_2' = \frac{2(n+1)}{(2n+1)} \tilde{\Theta}_2 = \frac{n+1}{2n+1} \chi_{\max}$$

$\Rightarrow M \tilde{\Theta}_2' = 0 \Rightarrow \tilde{\Theta}_2'$ - несмещ.

• дисперсия $\tilde{\Theta}_2$:

$$M \left[\frac{\chi_{\max}}{2} \right]^2 = \frac{n\theta^2}{4} \int_0^1 (1+t)^2 t^{n-1} dt$$

$$= \frac{n\theta^2}{4} \left(\frac{1}{n} + \frac{2}{n+1} + \frac{1}{n+2} \right) =$$

$$= \frac{\theta^2 (4n^2 + 8n + 2)}{4(n+1)(n+2)}$$

$$\begin{aligned} D \tilde{\Theta}_2 &= M \tilde{\Theta}_2'^2 - M^2 \tilde{\Theta}_2 = \frac{\theta^2 (4n^2 + 8n + 2)}{4(n+1)(n+2)} - \\ &- \frac{\theta^2 (4n^2 + 4n + 1)}{4(n+1)^2} = \frac{\theta^2}{4} \left(\frac{(4n^2 + 8n + 2)(n+1) - (4n^2 + 4n + 1)(n+2)}{(n+1)^2(n+2)} \right) = \\ &= \frac{n\theta^2}{4(n+1)^2(n+2)} \xrightarrow{n \rightarrow \infty} 0 \end{aligned}$$

$\Rightarrow \tilde{\Theta}_2$ - сф.

• дисперсия $\tilde{\Theta}_2'$:

$$D \tilde{\Theta}_2' = \frac{(2n+2)^2}{(2n+1)^2} D \tilde{\Theta}_2 \xrightarrow{n \rightarrow \infty} 0 \Rightarrow$$

$\tilde{\Theta}_2'$ - сф.

• Среднее значение ф.р. оценок:

$$\hat{\theta}_1 = \frac{2}{3} \bar{X}; \quad \hat{\theta}_2' = \frac{(n+1)}{2(n+1)} X_{\max}$$

$$D\hat{\theta}_1 = \frac{\sigma^2}{27n}$$

$$D\hat{\theta}_2' = \frac{n\sigma^2}{(2n+1)^2(n+2)}$$

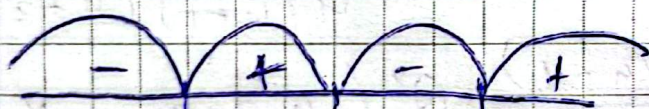
$$\frac{1}{27n} \sim \frac{n}{(2n+1)^2(n+2)}; \quad 27n^2 \sim (4n^2 + 4n + 1)(n+2)$$

$$4n^3 - 15n^2 + 3n + 2 \sim 0$$

$$(n-1)(4n^2 - 11n - 2) \sim 0, \quad 4n^2 - 11n - 2 = 0$$

$$n_{1,2} = \frac{11}{8} \pm \frac{3}{8}\sqrt{17} \Rightarrow n_1 \approx 3 \quad (\text{нужна})$$

$$n_2 \approx 0$$



$n_2 \quad n_3 \quad n_1$
0 1 3

нужна $n=3$

$$D\hat{\theta}_1' \geq D\hat{\theta}_2'$$

$\Rightarrow \hat{\theta}_1'$ эффективнее $\hat{\theta}_1$

тогда зададим функцию потерь для $\hat{\theta}$

$$f(\bar{X}_n, \theta) = \frac{X_{\max}}{\theta} - 1$$

$$P(f < t) = P(X_{\max} < \theta t + \theta)$$

$$= \left(P(\theta t + \theta) \right)^n = \left\{ f(x, \theta) = \right.$$

$$= \left(\frac{x}{\theta} - 1 \right) \{ \theta, 2\theta \} = (t+1-1)^n \{ \theta, 2\theta \} =$$

$$= t^* \{ (0; 2\theta) \}$$

$$t_1: f(t_1) = \frac{1-\beta}{2}$$

$$t_2: f(t_2) = \frac{1+\beta}{2}$$

$$\sqrt[n]{\frac{1-\beta}{2}} < \frac{x_{max}}{\theta} - 1 < \sqrt[n]{\frac{1+\beta}{2}}$$

$$P\left(\sqrt[n]{\frac{1-\beta}{2}} < \frac{x_{max}}{\theta} - 1 < \sqrt[n]{\frac{1+\beta}{2}}\right) = \beta$$

$$\sqrt[n]{\frac{1-\beta}{2}} + 1 < \frac{x_{max}}{\theta} < \sqrt[n]{\frac{1+\beta}{2}} + 1 \Rightarrow$$

$$\frac{x_{max}}{\sqrt[n]{\frac{1+\beta}{2}} + 1} < \theta < \frac{x_{max}}{\sqrt[n]{\frac{1-\beta}{2}} + 1}$$

• асимптотически эффективный план.

• метод непрерывных $\Rightarrow$ метод

выпуклости асимпт. эффективный ОММ.

$$\text{ОММ: } \tilde{\theta}_1 = \frac{2}{3}\alpha_1 = g(\tilde{\alpha}_1);$$

$$\nabla g(\alpha) = \frac{2}{3}; \quad \theta = g(\alpha)$$

$$K_{11} = \alpha_{i+1} - \alpha_i \alpha_i$$

$$\sqrt{n} \frac{g(\tilde{\alpha}) - g(\alpha)}{G(\tilde{\alpha})} \rightsquigarrow N(0; 1)$$

$$G^2(\alpha) = Dg^T(\alpha) K \cdot Dg(\alpha)$$

$$G^2(\alpha) = \frac{2}{3} (\alpha_2 - \alpha_1^2) \frac{2}{3} = \frac{4}{9} (\alpha_2 - \alpha_1^2)$$

$$\frac{\sqrt{n}(\hat{\theta}_1 - \theta)}{\frac{2}{3}\sqrt{\hat{x}_2 - \hat{x}_1^2}} \rightsquigarrow N(0; 1)$$

$$U \frac{1-\beta}{2} = t_1$$

$$\Rightarrow P\left(t_1 < \frac{\sqrt{n}(\hat{\theta}_1 - \theta)}{\frac{2}{3}\sqrt{\hat{x}_2 - \hat{x}_1^2}} < t_2\right)$$

$$U \frac{1+\beta}{2} = t_2$$

$$t_1 \cdot \frac{2}{3} \cdot \frac{\sqrt{\hat{x}_2 - \hat{x}_1^2}}{\sqrt{n}} < \hat{\theta}_1 - \theta < t_2 \cdot \frac{2}{3} \cdot \frac{\sqrt{\hat{x}_2 - \hat{x}_1^2}}{\sqrt{n}} \Rightarrow$$

$$\hat{\theta}_1 - \frac{2}{3} \cdot \frac{\sqrt{\hat{x}_2 - \hat{x}_1^2}}{\sqrt{n}} < \theta < \hat{\theta}_1 - \frac{2}{3} \cdot \frac{\sqrt{\hat{x}_2 - \hat{x}_1^2}}{\sqrt{n}} + 1$$

f), g), h) - аналогично предыдущим

$$6. \quad p(x) = \begin{cases} \frac{\theta-1}{x^\theta}, & x \geq 1 \\ 0, & x < 1, \end{cases} \quad \theta > 1.$$

$$a) \text{ ОМП: } L(\theta) = \frac{(\theta-1)^n}{\prod_{i=1}^n x_i^\theta} = \prod_{i=1}^n p(x_i, \theta)$$

$$\ln L(\theta) = n \cdot \ln(\theta - 1) - \theta \sum_{i=1}^n \ln x_i \Rightarrow$$

$$\frac{\partial \ln L(\theta)}{\partial \theta} = \frac{n}{\theta - 1} - \sum_{i=1}^n \ln x_i = 0 \Rightarrow$$

$$\frac{n}{\theta - 1} = \sum_{i=1}^n \ln x_i, \quad \frac{1}{n} \sum_{i=1}^n \ln x_i = \frac{1}{\theta - 1} \Rightarrow$$

$$\hat{\theta} = 1 + \frac{1}{\frac{1}{n} \sum_{i=1}^n \ln x_i}$$

$$\frac{\partial^2 \ln L(\theta)}{\partial \theta^2} = \frac{-n}{(\theta - 1)^2} < 0$$

$\forall \theta > 1 \Rightarrow$ максимум найден

б) упр. ① об асимпт. св. ОМР:

1) $\Xi = (1, +\infty)$ — область

2) условие регулярности:

$$\begin{aligned} & \text{регулярность} \cdot \frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} df(x, \theta) = \\ & = \int_{-\infty}^{+\infty} d \frac{\partial^2 f(x, \theta)}{\partial \theta^2} = \Rightarrow p(x, \theta) \in C^2 \end{aligned}$$

$p(x, \theta)$ — вып. функ. по θ на $(1, +\infty)$

$$\begin{aligned}
 \int_1^{+\infty} \frac{\partial p(x, \theta)}{\partial \theta} dx &= \int_1^{+\infty} \frac{\ln x - \theta \ln x + 1}{x^\theta} dx \\
 &= \frac{x^{1-\theta}}{1-\theta} \Big|_1^{+\infty} + \int_1^{+\infty} \ln x d(x^{1-\theta}) \\
 &= \frac{1}{\theta-1} - \frac{1}{\theta-1} = 0
 \end{aligned}$$

$$I(\theta) = E \left[\left(\frac{\partial \ln p(x, \theta)}{\partial \theta} \right)^2 \right] =$$

$$\begin{aligned}
 &= \int_1^{+\infty} \frac{\theta-1}{x^\theta} \left(\frac{1}{\theta-1} - \ln x \right)^2 dx = \\
 &= \int_1^{+\infty} \left(\frac{1}{x^\theta (\theta-1)} - 2 \frac{(\theta-1) \ln x}{x^\theta (\theta-1)^2} \right. \\
 &\quad \left. + \frac{(\theta-1) \ln^2 x}{x^\theta} \right) dx = \frac{1}{(\theta-1)^2} \\
 &\quad - 2 \int_1^{+\infty} \frac{\ln x}{x^\theta} dx + \int_1^{+\infty} \frac{(\theta-1) \ln^2 x}{x^\theta} dx = \\
 &= \frac{1}{(\theta-1)^2} + \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = \frac{1}{(\theta-1)^2} \\
 &\quad - \text{comp. na } (1; +\infty)
 \end{aligned}$$

$$I(\theta) > 0$$

- чужие параметры

$$\frac{\partial^2 f(x, \theta)}{\partial \theta^2} = \frac{-2 \ln x + \ln^2 x (\theta - 1)}{x^\theta}$$

$$\frac{\partial^2}{\partial \theta^2} \int_1^{+\infty} p(x, \theta) dx = 0$$

$$\int_1^{+\infty} \left(-\frac{2 \ln x}{x^\theta} + \frac{\ln^2 x (\theta - 1)}{x^\theta} \right) dx =$$

$$= \frac{2}{(\theta - 1)^2} - \frac{2}{(\theta - 1)^2} = 0$$

$p(x, \theta) = 2^\theta$ вып. функ.

на $\Theta = (1; +\infty)$ =) модель

чужие параметры

но $\oplus$ ОМР - ест.; асимпт.

норм., асимпт. норм., эксп.

асимптотич. норм.

$p = 0,5$ - частота

$$f(x_p) = p; \quad f(x_{\frac{1}{2}}) = 0,5$$

$$\int_1^x \frac{\theta - 1}{t^\theta} dt = \frac{-1}{x^{\theta-1}} + 1 = \frac{1}{2}$$

$$x_{\frac{1}{2}} = \frac{1}{2^{\theta-1}} \Rightarrow g(\hat{\theta}) = \frac{1}{2^{\hat{\theta}-1}}$$

$$g(\theta) = \frac{1}{2^{\theta-1}}$$

$$Dg(\tilde{\theta}) = 2^{\tilde{\theta}-1} \ln 2 \left(-\frac{1}{(\tilde{\theta}-1)^2} \right)$$

$$I^{-1}(\theta) = (\theta-1)^2 = 1$$

$$C^2(\theta) = \left(2 \cdot \frac{1}{\theta-1} \ln 2 \left(-\frac{1}{(\theta-1)^2} \right) \right)^2$$

$$(\theta-1)^2$$

$$\Rightarrow \frac{\sqrt{n} \left(2^{\frac{1}{\tilde{\theta}-1}} - 2^{\frac{1}{\theta-1}} \right)}{2^{\frac{1}{\tilde{\theta}-1}} \ln 2 \frac{1}{(\tilde{\theta}-1)}} \rightsquigarrow N(0; 1)$$

$$t_1 = \mathcal{U}_{\frac{1-\beta}{2}} ; t_2 = \mathcal{U}_{\frac{1+\beta}{2}}$$

$$P(t_1 < \frac{\sqrt{n}(\tilde{\theta}-1) \left(2^{\frac{1}{\tilde{\theta}-1}} - 2^{\frac{1}{\theta-1}} \right)}{2 \cdot \frac{1}{\tilde{\theta}-1} \ln 2} | t_0) =$$

$$= \beta$$

$$\frac{2^{\frac{1}{\tilde{\theta}-1}} \ln 2}{\sqrt{n}(\tilde{\theta}-1)} t_1 < 2^{\frac{1}{\tilde{\theta}-1}} - X \frac{1}{2} <$$

$$< \frac{2^{\frac{1}{\tilde{\theta}-1}} \ln 2}{\sqrt{n}(\tilde{\theta}-1)} t_2$$

$$2^{\frac{1}{\hat{\theta}-1}} - \frac{2^{\frac{1}{\hat{\theta}-1}} \ln 2}{\sqrt{n}(\hat{\theta}-1)} t_2 < X_{\frac{\alpha}{2}} < 2^{\frac{1}{\hat{\theta}-1}} -$$

$$- \frac{2^{\frac{1}{\hat{\theta}-1}} \ln 2}{\sqrt{n}(\hat{\theta}-1)} t_1$$

$$c) \sqrt{n} \frac{\tilde{\theta} - \theta}{g(\theta)} \rightsquigarrow N(0, 1)$$

$$g(\theta) = \theta \Rightarrow \nabla g(\theta) = 1$$

$$g^2(\theta) = 1 \cdot (\theta-1)^2 \cdot 1$$

$$\sqrt{n} \frac{\tilde{\theta} - \theta}{(\theta-1)} \rightsquigarrow N(0, 1)$$

$$\tilde{\theta} = 1 + \frac{1}{\frac{1}{n} \sum_{i=1}^n \ln x_i}$$

$$\sqrt{n} \cdot \frac{1}{n} \sum_{i=1}^n \ln x_i (\tilde{\theta} - \theta) \rightsquigarrow N(0, 1)$$

$$\Rightarrow (\tilde{\theta} - 1) = \frac{1}{\frac{1}{n} \sum_{i=1}^n \ln x_i} \cdot \frac{1}{\tilde{\theta} - 1} = \frac{1}{n} \sum_{i=1}^n \ln x_i$$

$$t_1 = n \frac{1-\beta}{2} ; t_2 = n \frac{1+\beta}{2}$$

$$P\left(t_1 < \frac{1}{\sqrt{n}} \sum_{i=1}^n \ln x_i (\tilde{\theta} - \theta) < t_2\right) = \beta$$

$$\tilde{\theta} - \frac{t_1}{\frac{1}{\sqrt{n}} \sum_{i=1}^n \ln x_i} < \theta < \tilde{\theta} - \frac{t_2}{\frac{1}{\sqrt{n}} \sum_{i=1}^n \ln x_i}$$

аккумулят. лит. габарит. не $\frac{1}{\sqrt{n}} \sum_{i=1}^n \ln x_i$